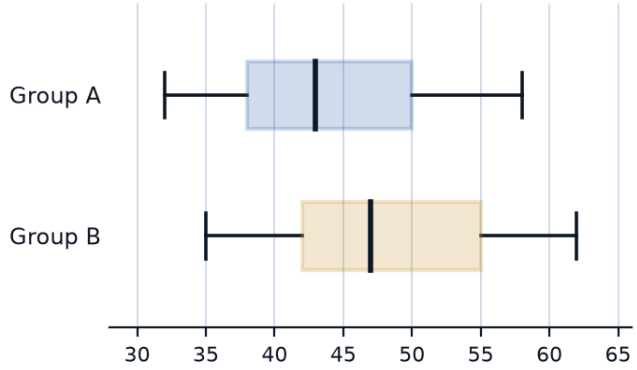
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the range of the data set: 20, 39, 10, 12, 26.

2 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the median of the data set: 32, 10, 29, 13, 37, 36, 34, 38.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Two independent events A and B have $P(A) = \frac{3}{4}$ and $P(B) = \frac{9}{10}$. Find $P(A \text{ and } B)$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A bag contains 5 red marbles, 4 blue marbles, and 3 green marbles. Two marbles are drawn at random one after another without replacement. What is the probability that both marbles are red?

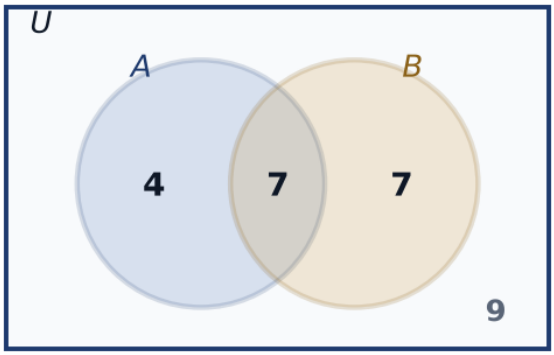
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The scores on a statewide standardized test are normally distributed with a mean of $\mu = 500$ and a standard deviation of $\sigma = 100$. Approximately what percentage of test takers score between 400 and 700?

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set contains the values: 4, 7, 7, 9, 12, 15. What are the mean, median, and range of this data set?



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A committee of 3 students is to be chosen from a group of 8 candidates. How many different committees can be formed?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A bivariate regression model for student preparation hours x and exam grade y yields the line of best fit $\hat{y} = 4.8x + 52$. A student who studied for 8 hours scored 94 on the exam. (a) What is the model's predicted score for this student? (b) Compute and interpret the student's regression residual.

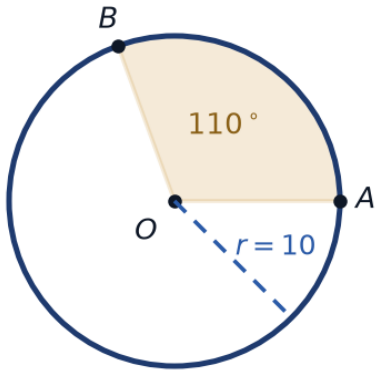
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A student has scores of 78, 85, 92, and 81 on four exams. What score must the student earn on the fifth exam to achieve an overall arithmetic mean of 85 across all five exams?

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10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The equation of a circle is given by $x^2 + y^2 + 10x - 4y + 20 = 0$. What is the radius of this circle?



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Each interior angle of a regular polygon measures 150°. How many sides does this polygon have?

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

An infinite geometric sequence has first term $a_1 = 18$ and common ratio $r = -\frac{1}{3}$. What is the sum of this infinite series?

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Which of the following expressions is equivalent to $\frac{1 - \sin^2\theta}{\cos\theta}$ for all values of θ where the expression is defined?

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

If $\cos\theta = -\frac{4}{5}$ and θ lies in Quadrant II, what is the value of $\tan\theta$?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

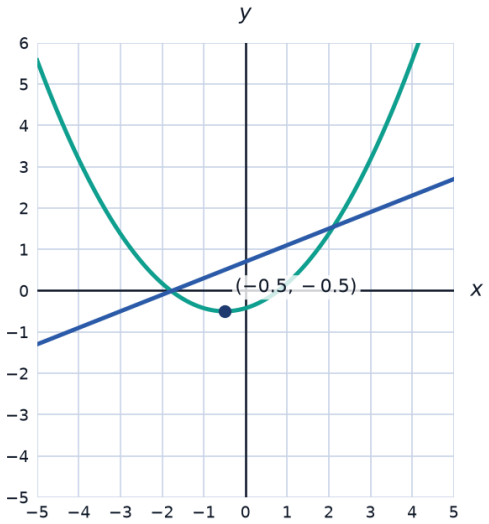
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

If $A = (2, -1; 3, 4)$ and $B = (5; -2)$, what is the matrix product AB?

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

Practice 1.1: A jacket costs 120 after a 25% discount. What was the original price before the discount?



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Line K passes through the points $(-3, 5)$ and $(1, -3)$. What is the slope of a line perpendicular to line K?

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Functions & Graphs

A right circular cylinder has radius 3 cm and height 8 cm. What is its total surface area in terms of π?

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Coordinate Geometry

What is the value of the complex expression $i^{47} - i^{18}$?

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Plane Geometry

What is the equation of the circle with centre $(-2, 4)$ and radius 5?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A student scored $82, 88, 91,$ and 85 on four exams. What score must the student earn on the fifth exam to achieve an overall mean score of 88?

a. 90

b. 92

c. 94

d. 96

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a set of 7 integers: $\{4, 7, 9, x, 14, 18, 20\}$ arranged in ascending order, the median is equal to the mean. What is the value of x?

a. 12

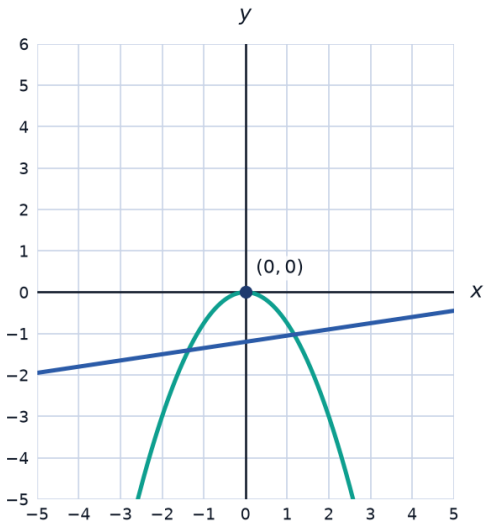
b. 13

c. 14

d. 15

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a chemistry course, homework counts for 20%, labs count for 30%, and tests count for 50%. If a student earns 90 on homework, 80 on labs, and 86 on tests, what is the weighted final grade?



a. 84

b. 85

c. 86

d. 87

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A data set consists of the numbers $\{3, 7, 8, 12, 15, 18, 21\}$. What is the interquartile range (IQR) of this data set?

a. 9

b. 11

c. 12

d. 14

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If every value in a data set is multiplied by 3 and then increased by 5, how is the standard deviation σ affected?

a. It increases by 5

b. It is multiplied by 3

c. It is multiplied by 3 and increased by 5

d. It remains unchanged

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which of the following data sets has a standard deviation equal to 0?

a. $\{5, 5, 5, 5, 5\}$

b. $\{1, 2, 3, 4, 5\}$

c. $\{-2, 0, 2\}$

d. $\{10, 20, 30\}$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A bag contains 6 red marbles, 8 blue marbles, and 10 green marbles. If one marble is selected at random, what is the probability that it is NOT blue?

a. $\frac{1}{3}$

b. $\frac{2}{3}$

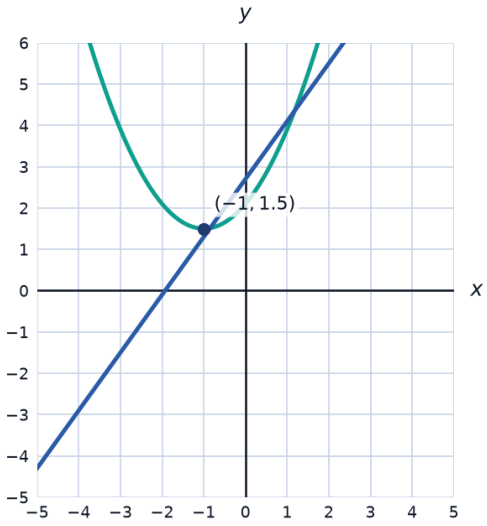
c. $\frac{1}{4}$

d. $\frac{3}{4}$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two fair standard six-sided dice are rolled simultaneously. What is the probability that the sum of the two numbers rolled is greater than 9?



a. $\frac{1}{6}$

b. $\frac{5}{36}$

c. $\frac{1}{9}$

d. $\frac{7}{36}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Events A and B are independent. If $P(A) = 0.4$ and $P(B) = 0.5$, what is the probability that at least one of the events occurs, $P(A \cup B)$?

a. 0.70

b. 0.80

c. 0.90

d. 0.20

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In how many distinct ways can a committee of 4 members be chosen from a group of 9 people?

a. 36

b. 84

c. 126

d. 3024

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How many different 4-digit identification codes can be formed using digits 0 through 9 if repetition of digits is not allowed and the first digit cannot be 0?

a. $4,536$

b. $5,040$

c. $9,000$

d. $3,024$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A restaurant offers 4 appetizers, 6 main courses, and 3 desserts. How many distinct 3-course dinner combinations are possible?

a. 13

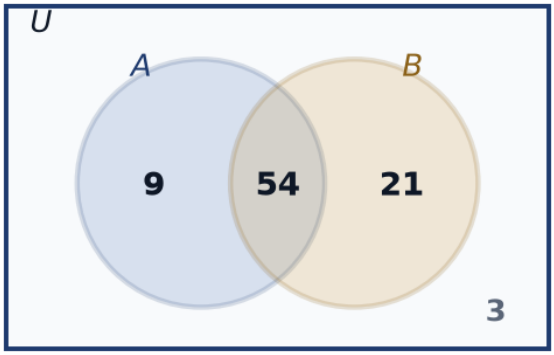
b. 36

c. 72

d. 144

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set of 9 values has a mean of 54. A new value of 224 is added. What is the new mean?



a. 43

b. 71

c. 47

d. 138

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34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A container holds only red, blue, and green marbles. Initially, the ratio of red marbles to blue marbles is $3:4$. First, 6 green marbles are added to the container, after which the probability of randomly drawing a green marble is $\frac{1}{3}$. Next, 4 red marbles are added to the container, after which the probability of randomly drawing a red marble is $\frac{2}{5}$. What was the total number of marbles originally in the container?

a. 14

b. 15

c. 21

d. 25

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A list of 5 positive integers has a mean of 12, a median of 10, and a unique mode of 8. What is the greatest possible integer that could be in this list?

a. 17

b. 22

c. 23

d. 24

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

A list consists of 7 distinct integers. The arithmetic mean of the 7 integers is 24, and the median is 22. If the 3 smallest integers are 15, 18, and 20, what is the maximum possible value of the largest integer in the list?

a. 46

b. 47

c. 48

d. 49

37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A bag contains 20 balls numbered 1 to 20. One ball is drawn at random. What is the probability that the number drawn is at most 3?

a. $\frac{14}{25}$

b. $\frac{1}{2}$

c. 0

d. $\frac{3}{20}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Functions & Graphs

A bag has 7 red, 8 blue, and 2 green marbles. One marble is drawn at random. What is the probability it is blue?

a. $\frac{3}{8}$

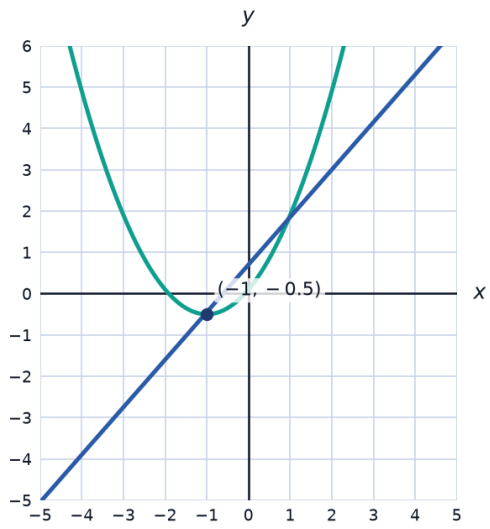
b. $\frac{1}{5}$

c. $\frac{1}{3}$

d. $\frac{8}{17}$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Coordinate Geometry

A bag has 3 red, 8 blue, and 3 green marbles. One marble is drawn at random. What is the probability it is red?



a. $\frac{13}{25}$

b. 1

c. $\frac{3}{16}$

d. $\frac{3}{14}$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Plane Geometry

A standard six-sided die is rolled once. What is the probability of rolling 3 or 4?

a. $\frac{15}{16}$

b. $\frac{1}{4}$

c. $\frac{1}{3}$

d. 0

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Trigonometry

A standard six-sided die is rolled once. What is the probability of rolling a number less than 4?

a. $\frac{1}{5}$

b. $\frac{3}{4}$

c. $\frac{7}{8}$

d. $\frac{1}{2}$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

A standard six-sided die is rolled once. What is the probability of rolling 1 or 2 or 4 or 5?

a. $\frac{2}{3}$

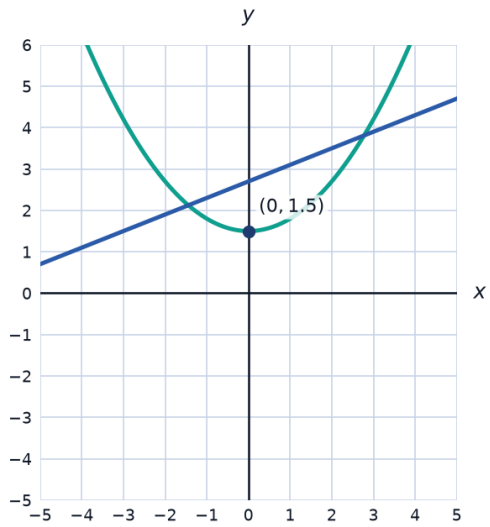
b. $\frac{9}{10}$

c. $\frac{3}{5}$

d. $\frac{1}{4}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

A class of 10 students averaged 62 on a test, and another class of 15 students averaged 77. What is the combined average?



a. 105

b. 158

c. 71

d. 122

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Find the mode of the data set: 9, 17, 5, 6, 9, 20.

a. 9

b. 2

c. 33

d. 37

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Functions & Graphs

If the probability of an event is $\frac{2}{15}$, what is the probability the event does NOT occur?

a. $\frac{13}{15}$

b. $\frac{9}{10}$

c. 0

d. $\frac{1}{8}$

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